

Human Models



Dylan Losey



This Lecture

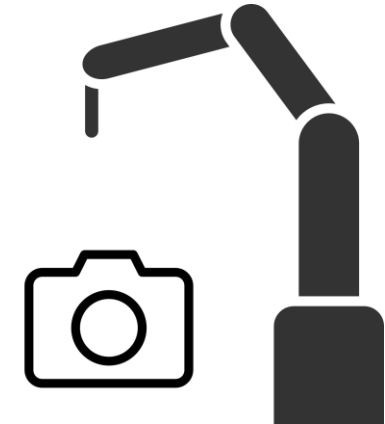


- How do we model humans?
- How do we use these models to learn from humans?

Learning from Humans

Robot observes the human's actions:

- *Tasks the human performs*
(showing an example trajectory)
- *Corrections the human provides*
(fixing the robot's motion)
- *Language inputs from the human*
(describing the task in words)
- *Human saying a motion is good or bad*
(binary feedback)



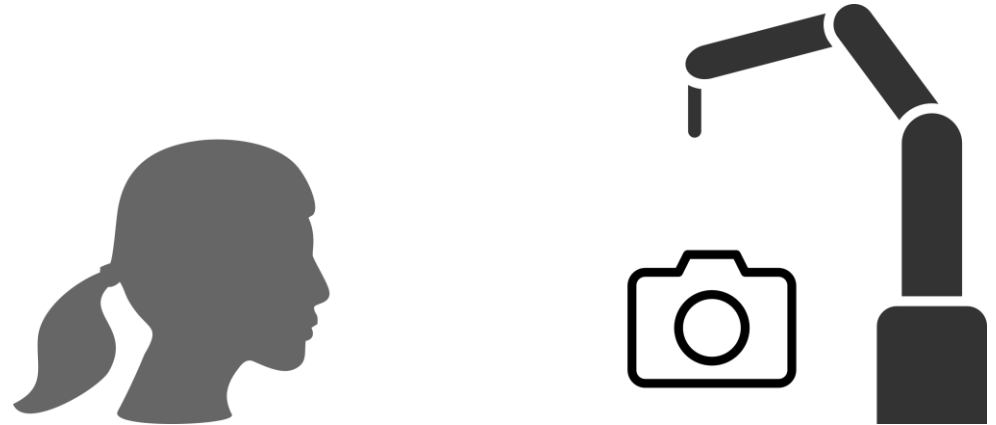
Robot needs to **interpret** these human actions,
i.e., find a meaning behind those actions

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To learn from the human's actions,
we use a **human model**

Learning from Humans

Imagine you are working with a human to build a block tower. For this problem setting, some intuitive **human models** could be:

1. *If the human reaches towards a block, they probably want to add that block to the tower*
2. *If the robot gets in the human's way, the human will probably wait for the robot to finish what it's doing before resuming their own motions*
3. *Humans generally like to build symmetric towers*
4. *Humans generally like to build taller towers*
5. *If the robot does a good job, the human will probably let the robot finish the tower by itself*



To understand
human models,
we must review
probability basics





id	class	MPS	PPV
0	vehicle.car		-
1	vehicle.bicycle		-
2	pedestrian.adult	cross.walk	-

Probability Basics

$$P(x)$$

Probability of x . Example: $P(\textit{turn_left})$

$$P(x, y)$$

Probability of x and y . Example: $P(\textit{turn_left}, \textit{slow_down})$

$$P(x \mid y)$$

Probability of x given y . Example: $P(\textit{slow_down} \mid \textit{stop_sign})$

Probability Basics

Two useful rules: **multiplication rule** and **marginalization**.

$$P(x, y) = P(x \mid y)P(y)$$

$$P(x) = \sum_{y \in Y} P(x, y)$$

Probability Basics

If x and y are two **independent** variables, then:

$$P(x, y) = P(x)P(y)$$

Use multiplication rule
where $P(x | y) = P(x)$

Example: $P(\text{turn_left}, \text{speeding}) = P(\text{turn_left})P(\text{speeding})$

Probability Basics

If x and y are **conditionally independent** given z , then:

$$P(x, y \mid z) = P(x \mid z)P(y \mid z)$$

Use multiplication rule
where $P(x \mid y, z) = P(x \mid z)$

Example: $P(\text{keys}, \text{wallet} \mid \text{drive}) = P(\text{keys} \mid \text{drive})P(\text{wallet} \mid \text{drive})$

Bayes Theorem

We often know how to model human actions **in one direction**:

If the human wants a block, they will likely reach for that block, $P(\mathbf{x} \mid \mathbf{y})$

But to infer what the human is thinking, we need to **reverse the model**:

If the human reaches for a block, what is their intent? $P(\mathbf{y} \mid \mathbf{x})$

Bayes Theorem

We can use Bayes Theorem to reverse a conditional probability:

$$P(y | x) = \frac{P(x | y)P(y)}{P(x)}$$

- $P(y)$ is the **prior**
- $P(x | y)$ is the **likelihood**
- $P(y | x)$ is the **posterior**
- $P(x)$ is the **normalizer**

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Practice problem: derive Bayes Theorem using the basic probability rules we have covered



Using models to learn from humans



DENSE CROWD



Core Idea

At their core, most problems that **learn from humans** are of this form:

$$P(y \mid x) \propto P(x \mid y)P(y)$$

$$P(\theta \mid s, a) \propto P(a \mid s, \theta)P(\theta)$$

Notation. Here s is the state, a is the human action, and θ is the task (or some other unknown parameters). We're trying to infer what task the human wants based on their actions

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Prior. What we know before any human actions.

Example: likely tasks are picking up blocks, stacking blocks...

Core Idea

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Human Model. Our assumption about how tasks (human's internal state) connect to actions.
Example: if human is holding a block, and they want to stack, they will move towards the tower...

Core Idea

At their core, most problems that **learn from humans** are of this form:

$$P(y \mid x) \propto P(x \mid y)P(y)$$

$$P(\theta \mid s, a) \propto P(a \mid s, \theta)P(\theta)$$

Posterior. What we have learned about θ after observing the human's action(s)

Example: because the human is carrying the block towards the tower, they likely want to stack...

Core Idea

Most works that learn from human-interaction make some **assumption or insight into the human's behavior**. This insight governs the human model:

$$P(a \mid s, \theta)$$

and connects the human's internal state θ (e.g., their goal, task, something we cannot directly observe) and the system state s to the observable human actions a



What are some
common human
models?



BOUNDED RATIONALITY

BOUNDED



UNBOUNDED



Bounded Rationality

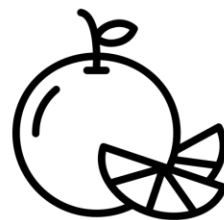
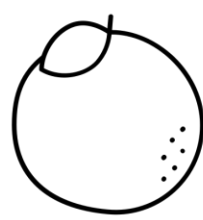
When humans perform tasks, we optimize for some internal objective. We can think of this as a scoring function: $r(a, s, \theta) \rightarrow \mathbb{R}$

when shopping, I want oranges that are fresh

A **rational** human will select actions to maximize $r(a, s, \theta)$

thoroughly check every orange buy the freshest one

$P(\text{buy})$



*A perfectly rational human
will sort through the bin and
always select this orange...*

Bounded Rationality

When humans perform tasks, we optimize for some internal objective. We can think of this as a scoring function: $r(a, s, \theta) \rightarrow \mathbb{R}$

when shopping, I want oranges that are fresh

A **bounded rational** human will select actions to *roughly* maximize $r(a, s, \theta)$

I'll check a few oranges and grab the best one I see

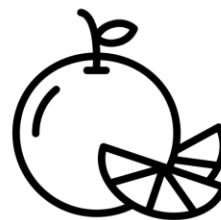
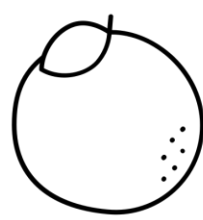
$P(\text{buy})$

Score +3

Score +5

Score +2

*A noisily rational human
chooses actions in proportion
to their score*



Bounded Rationality

To represent bounded rationality, we use this **human model**:

$$P(a \mid s, \theta) \propto e^{\beta \cdot r(s,a,\theta)}$$

β captures how **rational** the human is.

- When $\beta \rightarrow \infty$, the human always takes the optimal action (*perfectly rational*)
- When $\beta \rightarrow 0$, the human takes random actions (*uniform distribution*)

Bounded Rationality

For settings where the human has discrete choices, this becomes:

$$P(a \mid s, \theta) = \frac{e^{\beta \cdot r(s, a, \theta)}}{\sum_{a' \in A} e^{\beta \cdot r(s, a', \theta)}}$$

Note that $-\sum \log P(a \mid s, \theta)$ is the **cross-entropy loss**, a loss function commonly used in classification problems. Use this loss to learn parameters θ that best explain the given data

Next, the algorithm will query 10 preference questions:
Do you prefer Trajectory 0 or Trajectory 1?

Metropolis–Hastings

In practice, one of the key challenges for working with human models is computing probabilities over continuous variables. As a reminder:

$$P(\theta \mid s, a) = \frac{P(a \mid s, \theta)P(\theta)}{P(a)}$$

where $P(a)$ is a **normalizer**. It is not easy to compute $P(a)$ when θ is continuous. One way to sample θ from the posterior (*without computing the normalizer*) is Metropolis-Hastings

Metropolis–Hastings

$\theta \leftarrow P(\theta)$ sample some initial guess of the parameters

Overall goal: estimate θ from
observed human behavior

For each iteration i

$\theta' \leftarrow$ sample some new guess near θ

$P \leftarrow P(a | s, \theta)P(\theta)$ calculate unnormalized likelihood with θ

$P' \leftarrow P(a | s, \theta')P(\theta')$ calculate unnormalized likelihood with θ'

$\alpha \leftarrow P'/P$ since both have the same normalizer $P(a)$

if $\alpha > p$ where $p \sim U[0, 1]$:

$\theta \leftarrow \theta'$ update estimate of parameters

Return last value or last n values of θ

Use your human model to
calculate these probabilities.
Intuitively, we want to compare
them and see whether the human
action is more likely under θ or θ'

Metropolis–Hastings

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Can think of this as stochastic gradient ascent. We always take θ' if it explains the data better than θ , and we sometimes take it even if it is worse (α is the acceptance rate)

Takeaways

- We observe human actions, and we want to infer hidden information
- Human model connects the hidden information to the actions
- General form of learning from humans:

$$P(\theta \mid s, a) \propto P(a \mid s, \theta)P(\theta)$$

- We can improve learning by producing better **human models** $P(a \mid s, \theta)$ or by reaching a better understanding of the **prior** $P(\theta)$

This Lecture



- How do we model humans?
- How do we use these models to learn from humans?

Next Lecture



- Using human models to learn in real-time and share autonomy